



BRILL

HL Mandarin Speakers Toss the Same Way as Fluent Mandarin Speakers

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Abstract

Heritage language (HL) speakers often show weaker semantics in HL words than speakers who continue to learn and use the language. In this study, we tested whether HL Mandarin speakers simplified near-synonyms of throw verbs by diminishing the difference between the near-synonyms and/or by diminishing the difference between the generic throw verb and other near-synonyms. Two participant groups, HL Mandarin speakers and English second-language learners, acted out six Mandarin near-synonyms of throw verbs and the English verb *throw*. The results showed more similarities than differences between the two groups in the core features of throw verb semantics (force, speed, and direction). We observed few signs of simplification. One interpretation of these results is that early and/or naturalistic exposure to Mandarin was an important predictor of speakers' knowledge of conceptual features. These results add to the literature showing that there can be factors beyond proficiency that contribute to speakers' lexical semantics.

Keywords

bilingual acquisition – concept formation – second language acquisition – heritage language – lexical semantics

1 Introduction

In communities in which there is a majority language, many first-language speakers of minority languages become heritage language (HL) speakers (Montrul, 2016, 2010; Scontras et al., 2015). HL speakers typically acquire their HL as children in the home environment, learning it from family members as one of their primary languages (Scontras et al., 2015). HL speakers often start speaking the majority language more often (even with family members) after they start attending school in the majority language (Bridges & Hoff, 2014; Jia et al., 2014). As a result, as adults, HL speakers are often weaker in proficiency in their first language than in their second (Jia et al., 2014; Kagan, 2005; Mu, 2014; Polinsky, 2011; Potowski, 2004). Also, HL speakers tend to use their HL differently from both monolingual speakers of that language and bilinguals who maintain a high degree of proficiency in both languages (Schmid & Köpke, 2017; Scontras et al., 2015; Treffers-Daller & Tidball, 2015). For example, HL speakers are often highly variable in producing target-like linguistic constructions (Karayayla, 2020), because their HL knowledge is attriting (Polinsky, 2011; Schmid &

Karayayla, 2020) and/or because they never fully acquired the constructions before their second language became dominant (Montrul, 2010, 2016, 2018).

Some studies have suggested that HL speakers' language use is characterized by simplification. Simplification refers to speakers' tendency to streamline linguistic structures in order to make them easier to process (Silva-Corvalán, 1986). Simplification can involve the reduction of morphological complexity or the application of simpler syntactic structures or more semantically generic word choice (Pinto & Raschio, 2007; Treffers-Daller & Tidball, 2015). Some previous studies have reported signs of simplification among HL speakers (Montrul, 2010, 2015, 2018). For example, Gao (2010, 2015) found that both English-Chinese bilingual children and adults overused the generic classifier *ge* 个 in Chinese, where monolinguals would use more specific classifiers. Similarly, Lin and Nicoladis (2018) showed that, when telling the story of motion events that had happened in a cartoon, 8–10-year-old HL Mandarin speakers used more single deictic verbs *lái lái* 来来 'come-here' and *qù qù* 去去 'go-there' than either monolingual children or younger bilingual children. The older bilinguals expressed on average 32% of their motion events with single deictic verbs compared to closer to 10% for all other groups. In contrast, the other children used more precise motion-event terms. Not all studies have found evidence for simplification among HL speakers in the domain of semantics (Chamorro, Sturt, & Sorace, 2016; Montrul, 2006). The objective of the present study was to test whether HL speakers of Mandarin would show signs of simplification of the meaning of near-synonyms for throw verbs.

Near-synonyms refer to words sharing similar meanings but conveying differing concepts or emotions, or words that are used in different linguistic environments (Gao & Wang, 2012). For example, in English, *burgundy*, *cranberry*, *claret*, and *rust* can all refer to shades of red. Similarly, *toss*, *hurl*, *fling*, and *cast* are near-synonyms of throw verbs in English. In Mandarin, near-synonyms of throw verbs differ on semantic dimensions of force, speed, direction, and whether the action has a path or event endpoint (Gao, 2001a; Gao & Wang, 2012; Gao, Wang & Nicoladis, 2016; Lui et al., 2000; Nicoladis & Gao, 2021; Wang & Gao, 2016). For example, the throw verb *diū* 丢 is used to describe random throwing with an event endpoint involving no specific path or direction. In contrast, the verb *rēng* 扔, which also describes recklessly throwing something away, does not involve a path or event endpoint or specify a direction of motion, and is the most generic throw verb, largely equivalent to throw in English (Gao & Wang, 2012; Nicoladis & Gao, 2021). If the HL speakers were simplifying near-synonyms for throw in Mandarin, one hypothesis is that they would not differentiate between the force, speed, and direction of the generic verb *rēng* 扔 (on the one hand) and other near-synonyms for throw (on the

other hand). Additionally, the HL speakers might demonstrate minimal variation in force, speed, and direction with other near-synonyms, essentially treating them all as close synonyms.

We compared the HL speakers' conceptualization of Mandarin near-synonyms for throw to that of Mandarin speakers who spoke English as a second language (ESL speakers). All the participants acquired Mandarin as a first language at home. The HL speakers became English dominant speakers as they started schooling in English in Canada, while the ESL speakers started schooling in China and moved to Canada for higher education. The dominant language of the ESL speakers remained Mandarin. We invited our participants to enact six Mandarin throw verbs, *rēng* 扔, *diū* 丢, *pāo* 抛, *tóu* 投, *shuāi* 摔, *shuǎi* 甩, and the English verb throw as the stimuli in this study. The differences in meaning of these throw verbs are evident in the action contours of each one. All six throw verbs are acquired by L1 speakers at an early age, as they are commonly used basic physical action verbs (Gao, 2001a). According to Gao's (2001a) corpus-based semantic decomposition, there are many conceptual features that distinguish throw verbs, including force, speed, direction, body part, duration, agent's intention for the action, agent's predicted causal effect of the action, possible end point, and properties of the object to be thrown (e.g., small, hand size, or bigger), etc.

The foundation of Gao's (2001a) work is the theoretical framework of lexical decomposition in the field of linguistics. Lexical decomposition posits that understanding the meaning of a complex expression, such as a word or phrase, is achievable by breaking it down into smaller, more fundamental semantic components or features. The methodology involves a thorough examination of the constituent elements of a linguistic unit to analyze its meaning. Within this framework, complex expressions are conceptualized as possessing an internal structure composed of simpler, atomic semantic features. These features act as the fundamental building blocks of meaning, and their amalgamation contributes to the comprehensive interpretation of the expression. Decompositional semantics asserts that understanding the meaning of a word or phrase is best achieved through a detailed examination of its constituent parts, rather than treating it as a single, indivisible unit. Several notable scholars have made significant contributions to lexical decomposition, including Ray Jackendoff (2002, 2003, 2007), Jerry A. Fodor (2001a, 2001b, 2013), Eleanor Rosch (1976, 2008), George Lakoff (1973, 1986), and Raymond W. Gibbs Jr. (1984, 1989, 2003). The overarching objective is to explore how the meanings of complex expressions are derived from more basic semantic components.

In the domain of throw verbs, there are many semantic features that can be discerned from the lexical meanings (Gao, 2001a). In this study, we focus on the three features that are key in distinguishing near-synonyms. Following

previous research (Levin, 1993; Talmy, 2000, 2016; Viberg, 2004, 2016), we focus on the following three core semantic features (Gao, 2001a, 2001b):

- Force: The exertion of physical strength or energy involved in propelling an object through the air.
- Speed: The rate at which the object is propelled through the air.
- Direction: The intended path or trajectory of the thrown object.

By breaking down the meaning of throw into these fundamental semantic components, we gain a more nuanced understanding of the action. Each component contributes to the overall interpretation of the verb:

- The force applied gives insight into the intensity or effort involved in the throwing action.
- The direction provides information about where the object is aimed or intended to go.
- The speed conveys the velocity at which the object travels through the air.

In a decompositional analysis, these features serve as the elemental building blocks, allowing us to grasp the intricacies of throw verbs by examining their constituent parts.

1.1 *Enactment Methodology*

In this study, we invited participants to enact the throw verbs. By inviting participants to enact the meaning of these verbs, we can tap into their semantic knowledge of those words without the influence of surrounding words. This method made it possible to pinpoint the core elements – force, speed, and direction – within the conceptual semantics of throw verbs. Previous studies have used this methodology to study the core semantic features of throw verbs among monolingual speakers of Mandarin, English, and German (Gao & Wang, 2012; Gao et al., 2016). These studies showed that monolingual Mandarin speakers differentiate all six of the near-synonyms included in this study in terms of their force, speed, and direction. Moreover, there was high inter-participant agreement among these three core semantic features for each verb. The results also showed that the generic Mandarin verb *rēng* 扔 was similar to the generic English verb throw in terms of force, speed, and direction.

2 This Study

The purpose of this study was to test whether HL speakers would simplify near-synonyms of throw verbs in their first and weaker language, Mandarin, and whether English as a Second Language (ESL) speakers, whose dominant language is Mandarin, would make more conceptualized distinctions between

the near-synonyms. We explored these questions by inviting the two groups of participants to perform an enactment task.

Based on previous studies, we made the following hypotheses: First, since HL speakers have weaker access to Mandarin than ESL speakers, we hypothesized that they might simplify or overgeneralize the conceptual differences between the near-synonyms, and thus making the near-synonyms less differentiated. If that were the case, ESL speakers would demonstrate discernible variations in the force, velocity, and trajectory of the six Mandarin near-synonyms, whereas HL speakers would not exhibit such distinctions. Additionally, or alternatively, HL speakers might generalize the core semantics of the generic throw verb *rēng* 扔 to all Mandarin near-synonyms. In this scenario, HL speakers might not exhibit variations in the force, speed, and direction of *rēng* 扔 and the other five Mandarin near-synonyms. In contrast, ESL speakers would distinguish all six verbs.

We also compared the participants' enactments of the Mandarin near-synonyms with the highly frequent English verb throw. It is possible that HL speakers generalize the meaning of this word from their dominant language to the meaning of the Mandarin throw verbs. As we will discuss more in detail in the Discussion, this comparison is not a strong test of whether the HL speakers are transferring their knowledge from English to Mandarin.

3 Method

3.1 Participants

A total of 49 adult Mandarin-English bilinguals ($M = 21.1$ years old, $SD = 3.3$) were recruited in Edmonton, Canada for this study. In this region of Canada, English is the majority language. Participants completed a questionnaire before beginning the experiment to assess their proficiency and general language background in English and Mandarin. All participants learned to speak Mandarin at home as their first language and started to learn and speak English when they started their schooling. Twenty-four of them identified English as at least one of their dominant languages (i.e., two HL speakers reported that they had balanced proficiency in both languages). Their mean experience with English was 15.1 years, $SD = 3.0$ years. Twenty-five of the participants identified Mandarin as their dominant language and were ESL speakers. Their mean experience with English was 12.0 years, $SD = 4.5$ years. Based on the participants' proficiency in English and Mandarin, we divided them into two groups: HL speakers, with Mandarin as their first language and English as a dominant language, and ESL speakers, with Mandarin as their first and dominant language and English as their second and weaker language.

All participants learned English through schooling. For the HL speakers, English was the main language of instruction of school in Canada, while the ESL speakers learned English as a school subject in Mainland China. The HL speakers reported that they still heard and sometimes used Mandarin at home but used English in the majority of their interactions. Unsurprisingly, the HL speakers had significantly longer experience with English than the ESL speakers, $t(47) = 2.83$, $p = .0007$, $d = .812$.

Participants completed two language sessions (see section 3.3) and received a Cdn\$10 honorarium for each one, totaling Cdn\$20 after the two sessions.

3.2 *Materials*

Three objects, a stuffed animal dolphin, a soft football, and a hard orange ball, were given to the participants for them to choose one in performing the enactments. The verbs to be acted out in the task were shown on white index cards. They were the six throw verbs in Mandarin: *rēng* 扔, *diū* 丢, *pāo* 抛, *tóu* 投, *shuāi* 摔, *shuǎi* 甩. We interspersed the six throw verbs with six verbs referring to picking up or grabbing, in order to ensure that the participants knew that the target verbs referred to throwing. To test if HL speakers were transferring conceptual features from enactments of their dominant language to pantomimes of their weaker language, the English throw verb throw was added to the English session for HL speakers only.

The standardized receptive vocabulary task, the Peabody Picture Vocabulary Test-III Edition (PPVT; Dunn & Dunn, 1997), was included for the assessment of the participants' proficiency in English. To assess the participants' working memory capacity, two standardized tests from the Automated Working Memory Assessment (AWMA; Alloway, 2007) were included. One was the listening recall task to examine participants' verbal working memory, and the other was the spatial recall task to examine participants' visuospatial working memory.

3.3 *Procedure*

For the PPVT test, the experimenter presented the participant with an array of four pictures, each labeled with a number from one to four, and read out a word. Participants were asked to indicate the picture number which best represented the word. The responses were scored as either correct or incorrect.

In the two standardized memory tests, first was a listening recall task, followed by a spatial recall task. The computer program provided participants with instructions for completing the task, which were supplemented or clarified by the experimenter when necessary. The experimenter judged the accuracy of each response using an answer key, to control the progression of the task. Participants heard sentences and were asked to report if the statement was true or false (processing score) and then recite the last word of each

sentence in the same order the sentence(s) were presented (memory score). Participants heard an increasing number of sentences in each block and could hear a maximum of seven sentences in one block.

In the spatial recall task, participants were presented with two images. One served as a reference object, and the other was a rotated object. Participants had to determine whether the rotated object was the same or opposite to the reference object (processing score). Afterward, they were asked to indicate the location(s) of the red dot that appeared at the end of each rotated object, recalling the order in which they saw the objects (memory score). The number of paired objects shown gradually increased in each block, with participants dealing with up to seven paired objects.

For the enactment task, participants were tested in two separate sessions (one in Mandarin and one in English), scheduled about a week apart. The order of these sessions was counterbalanced across participants. In the Mandarin session, participants were videotaped as they completed the enactment task in Mandarin. The experimenter spoke Mandarin when administering the task. The experimenter presented participants the index cards with pinyin spelling and Chinese characters for each verb. The order of presentation was randomized for each participant, interspersing the *grab/touch* verbs and the throw verbs. The experimenter pronounced the verb. Participants were asked to act out the verb using any of the three objects (dolphin, football, and ball) provided. After participants acted out each verb once, the experimenter shuffled the cards, and participants acted out the verbs a second time. If participants did not understand a verb presented, they were told that they could clearly speak to the camera indicating they did not know it, and they could either guess an action or pass. In the experiment, there were no such cases; all participants responded and acted out all the throw verbs. The Mandarin session took approximately 15 minutes.

In the English session, participants were interviewed by an experimenter who was a native speaker of English. Participants completed the PPVT. Then, participants completed two tests from the AWMA on the computer, administered by the experimenter. First participants completed the listening recall task. They had the choice to take a two-minute break following the completion of this task or could proceed with the spatial recall task. At the end, HL speakers were asked to act out the English verb throw. The instruction was the same as given in the Mandarin session. The English session took approximately 50 minutes.

3.4 *Coding Enactments*

Participants' enactments were coded based on the following dimensions: force, speed, wrist movement, wrist form, type of throw, direction of throw, repetition of action, rotating the body, number of fingers used, and use of the palm of the hand (these variables distinguish near-synonyms both within and between languages; Gao & Wang, 2012; Wang & Gao, 2012; Gao et al., 2016). Force was coded out of 3 with 1 = *little force*, 2 = *medium force*, and 3 = *great force*. Speed was coded out of 3 with 1 = *slow motion*, 2 = *medium speed*, 3 = *fast motion*. The direction of throw was coded as sideways, forward, upward, downward, circular, and backward. A sideways motion involved the participant's arm, hand, and/or the object moving across their body. A forward motion involved the participant's arm, hand, and/or the object advancing in front of their body. An upward motion involved the participant's arm, hand, and/or object making an ascending motion. A downward motion involved the participant's arm, hand, and/or object making a descending motion resulting in the object making contact with the ground. A circular motion involved the participant twirling their forearm, hand, and/or object, as if drawing a circle. A backward motion involved the participant's arm, hand, and/or object advancing behind their body. The repetition of action and the use of the palm of the hand (where the object made contact with the palm of the hand during the throw) were coded as being present or not. The number of fingers used was counted with five fingers, indicating the use of one hand, and ten fingers, indicating the use of two hands. We present the results only of the force, speed, and direction, the core semantic features of our target words.

3.5 *Inter-rater Reliability*

We asked two coders to independently code all 300 videotaped throw events. Both coders were students majoring in psychology and engaged in psycholinguistic research. In order to perform the coding with no influence from knowledge of Mandarin, neither spoke any Mandarin at all. We coded as *agreed* when both coders gave the same response (including for no responses). For force, the two coders gave the same rating to 228 of the events (76%). For speed, they gave the same rating to 187 events (62%). They gave the same directional coding for 242 events (81%). We asked a third coder to independently code for the 243 (out of 900) coding categories that the two coders did not agree on. In all but 5 instances, the third coder agreed with one coder or the other, so we included in the analysis the number that two out of three coders agreed on. In the five instances of disagreement, two were direction and we excluded those from the analyses. The remaining three were on numerical scales (one

force and two speed), so we averaged all three coders' codes to include in the analyses.

4 Results

4.1 English Proficiency and Working Memory

The average raw scores of the test results are presented in Table 1. As expected, the HL speakers scored significantly higher on the PPVT than the ESL speakers, $t(45) = 8.89, p < .001, d = 2.622$. The HL speakers also scored significantly higher than the ESL speakers on both the verbal memory processing score, $t(43) = 4.36, p < .001, d = 1.301$, and on the verbal memory capacity score, $t(43) = 4.34, p < .001, d = 1.295$. Verbal working memory has been linked with language knowledge (Rose & Rouhani, 2012). The correlation between PPVT scores and performance on the verbal memory measures were positive and significant, both for processing, $r(42) = .654, p < .00001$, and for capacity, $r(42) = .626, p < .00001$. So it was not surprising to observe that ESL speakers scored lower on these tasks than HL speakers.

There were no significant differences between the two groups on either the visuospatial memory processing score, $t(43) = 0.47, p = .64, d = .139$, or on the visuospatial memory capacity score, $t(43) = 0.19, p = .85, d = .058$.

TABLE 1 Average (SD) on background measures for ESL and HL speakers

	ESL speakers	HL speakers
PPVT	158.4 (9.5)	181.3 (7.5)
Verbal memory: processing	19.3 (8.2)	32.6 (12.0)
Verbal memory: total score	12.3 (3.3)	16.7 (3.5)
Visuospatial memory: processing	102.0 (40.4)	96.8 (34.1)
Visuospatial memory: total score	26.9 (7.2)	26.5 (5.4)

4.2 Force

We first compared how HL speakers and ESL speakers acted out Mandarin near-synonym throw verbs based on force, with a 2×5 [Group $\times$ Verb] ANOVA, with Verb as a repeated measure. This analysis revealed a significant main effect for Verb, $F(5, 85) = 11.23, p < .001, \eta^2 = .392$, but no main effect for Group, $F(1, 17) = 0.01, p = .93, \eta^2 = .001$, and no interaction effect, $F(5, 85) = 0.85, p = .84, \eta^2 = .014$. In other words, the participants threw with different levels of force

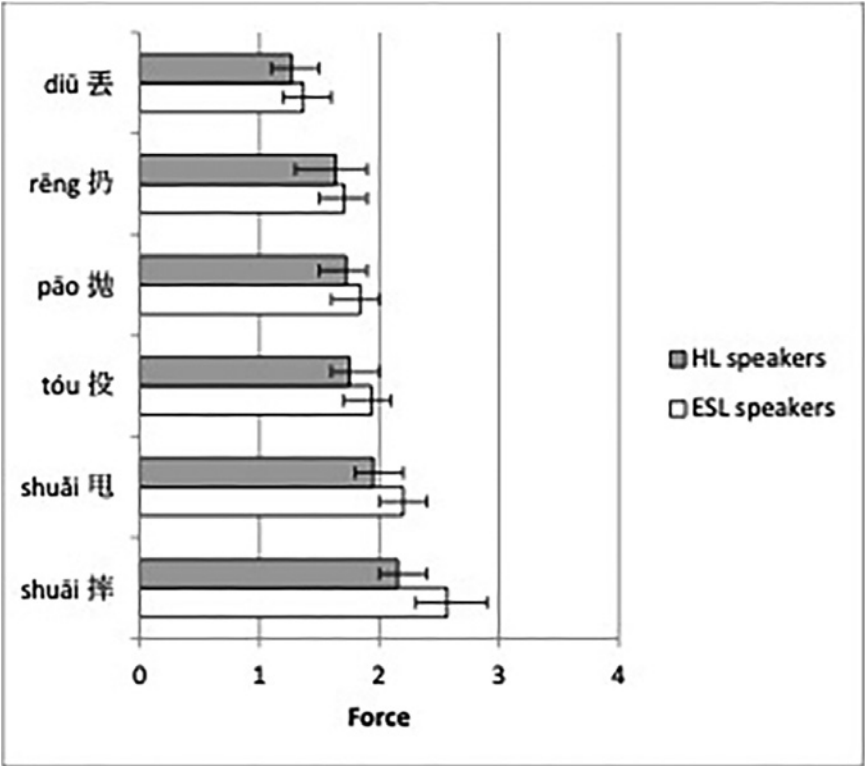


FIGURE 1 Average force for each verb. Note: error bars indicate bootstrapped 95% confidence intervals

for different verbs (see Appendix 1 for post-hoc tests for each verb), but there was no overall difference between HL speakers and ESL speakers on the force used to act out each near-synonym, as shown in Fig. 1.

We did a follow-up analysis comparing the two groups on the force for each verb using planned independent-samples t-tests. The difference between the two groups reached significance for *shuāi* 摔, $t(40) = 2.07, p = .045, d = .640$, but no other differences reached significance ($ps > .15$). As can be seen in Fig. 1, the HL speakers threw with less force for *shuāi* 摔 than ESL speakers. While this difference was statistically significant, it is important to note that, like the ESL speakers, the HL speakers threw with more force for *shuāi* 摔 than for any other near-synonym.

4.3 Speed

Figure 2 summarizes the speed with which the participants threw for each near-synonym. We compared how HL speakers and ESL speakers acted out

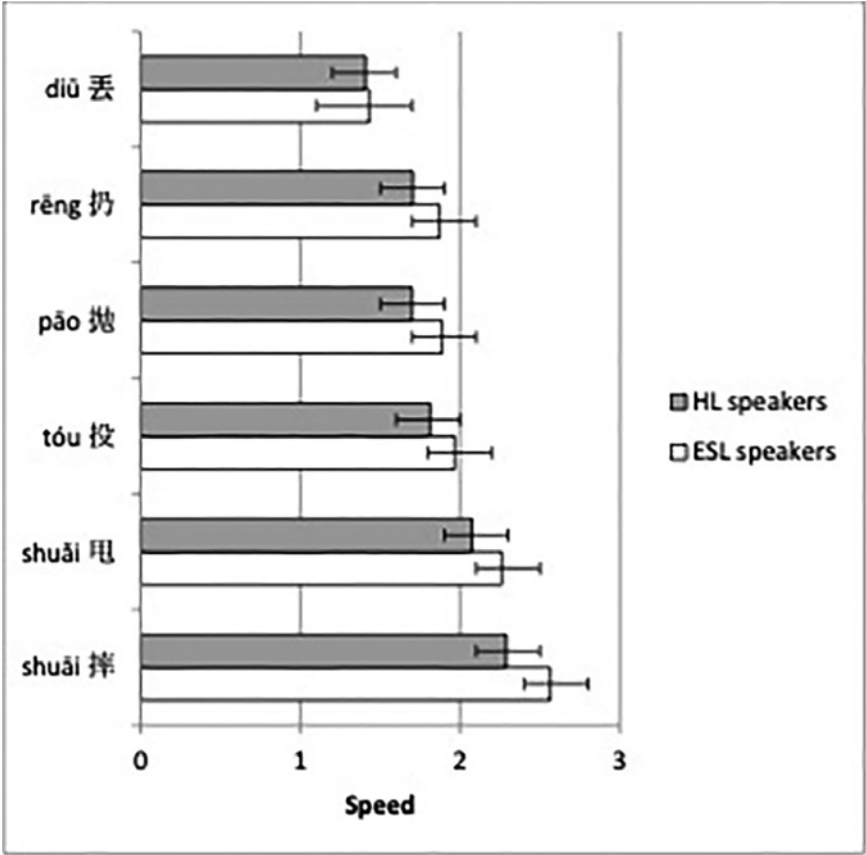


FIGURE 2 Average speed for each verb. Note: error bars indicate bootstrapped 95% confidence intervals

Mandarin near-synonym throw verbs based on speed with a 2 x 5 [Group x Verb] ANOVA, with Verb as a repeated measure. This analysis revealed a significant main effect for Verb, $F(5, 85) = 10.67, p < .001, \eta^2 = .381$, but no main effect for Group, $F(1, 17) = 0.01, p = .92, \eta^2 = .001$, and no interaction effect, $F(5, 85) = 0.33, p = .89, \eta^2 = .012$. As shown in Fig. 2, these results suggest that the participants differentiated the near-synonyms with respect to speed, but there was no significant difference between the HL speakers and the ESL speakers in how quickly they threw for the near-synonyms. To verify the result, we ran independent-samples t-tests, comparing the two groups for each verb. These analyses revealed no significant differences between the groups, $ps > .14$.

4.4 Direction

To see if there were any differences between the two groups in the direction that they acted out the throwing, we counted the number of people in each

group who threw in each direction (see Appendix 1). We then compared these numbers using chi-square tests. These analyses revealed no significant differences between groups, for *rēng* 扔, χ^2 ($df = 3$, $N = 43$) = 6.22, $p = .10$, *diū* 丢, χ^2 ($df = 3$, $N = 44$) = 2.25, $p = .52$, *pāo* 抛, χ^2 ($df = 3$, $N = 42$) = 3.88, $p = .28$, *tóu* 投, χ^2 ($df = 1$, $N = 29$) = 0.36, $p = .55$, *shuāi* 摔, χ^2 ($df = 1$, $N = 41$) = 1.19, $p = .28$, and *shuǎi* 甩, χ^2 ($df = 5$, $N = 43$) = 5.08, $p = .41$.

For most of the near-synonyms, there were only one or two preferred directions of throwing in both groups. For *rēng* 扔, both HL speakers and ESL speakers preferred forward movements (65% of participants threw in a forward direction). For *diū* 丢, most HL speakers inclined downward (46%) or forward throwing (36%). ESL speakers also showed a tendency to act out downward throwing but were more scattered in their direction choices than HL speakers. For *pāo* 抛, the majority of the participants from both groups threw in an upward direction (64%). For *tóu* 投, the majority of participants from both groups threw in a forward direction (59%), though many threw in an upward direction (41%). For *shuāi* 摔, all but one participant threw downward (i.e., 98%).

For *shuǎi* 甩, there was no clear preference for direction for either ESL speakers or HL speakers. Participants from both groups demonstrated this verb using various directions, including sideways, forward, upward, downward, circular, and backward.

4.5 Differences from *rēng* 扔

As a basic or generic verb of throwing, *rēng* 扔 has the most representative action contour of throw verbs. Did the enactment of *rēng* 扔 differ from that of its five near-synonyms? Table 2 summarizes the results. The significant differences compared with *rēng* 扔 in ESL speakers' pantomimes were found in force and speed of *diū* 丢, all three conceptual features in *shuāi* 摔, and force and speed in *shuǎi* 甩, while in HL speakers' pantomimes only force and speed of *shuāi* 摔 showed significant differences. The differences are labeled by the letter Y (= yes) and N (= no) and highlighted in Table 3.

4.6 Chi-square Analyses

To test if the participants distinguished each near-synonym from *rēng* 扔, we coded whether each response for each characteristic was the same as *rēng* 扔 (1) or different (0). We then compared the two language groups on a series of chi-square analyses. In terms of force, there was no difference between language groups ($ps > .32$) nor speed ($ps > .25$). In terms of direction, there was one significant difference, for *pāo* 抛, χ^2 ($df = 1$, $N = 34$) = 7.54, $p = .006$. Most HL speakers produced *pāo* 抛 in the same direction as *rēng* 扔, but most ESL speakers did not. All other differences were not significant, with $ps > .33$.

TABLE 2 *Rēng* 扔 compared with its five near-synonyms

Near-synonyms of <i>rēng</i> 扔	Conceptual feature	ESL speakers	HL speakers
<i>diū</i> 丢	Force	$t(17) = 2.19, p = .04$	$t(19) = 1.67, p = .11$
	Speed	$t(17) = 2.82, p = .01$	$t(19) = 1.44, p = .17$
	Direction	$\chi^2(df = 6, N = 19) = 3.86, p = .70$	$\chi^2(df = 1, N = 17) = 6.36, p = .10$
<i>pāo</i> 抛	Force	$t(16) = 0.30, p = .77$	$t(18) = 0.71, p = .49$
	Speed	$t(16) = 0.48, p = .64$	$t(18) = 0.33, p = .75$
	Direction	$\chi^2(df = 4, N = 19) = 4.44, p = .35$	$\chi^2(df = 2, N = 15) = 2.25, p = .33$
<i>shuāi</i> 摔	Force	$t(16) = 3.92, p = .001$	$t(16) = 2.43, p = .03$
	Speed	$t(16) = 3.43, p = .003$	$t(16) = 2.50, p = .02$
	Direction	$\chi^2(df = 1, N = 17) = 11.41, p = .003$	$\chi^2(df = 1, N = 17) = 2.55, p = .11$
<i>shuāi</i> 甩	Force	$t(18) = 3.16, p = .005$	$t(18) = 1.68, p = .11$
	Speed	$t(18) = 2.88, p = .01$	$t(18) = 1.56, p = .14$
	Direction	$\chi^2(df = 8, N = 19) = 12.61, p = .13$	$\chi^2(df = 2, N = 16) = 3.77, p = .44$
<i>tóu</i> 投	Force	$t(11) = 1.73, p = .11$	$t(15) = 0.44, p = .67$
	Speed	$t(11) = 1.30, p = .22$	$t(15) = 0.64, p = .53$
	Direction	$\chi^2(df = 1, N = 14) = 1.05, p = .59$	$\chi^2(df = 1, N = 14) = 0.01, p = .91$

TABLE 3 Highlight of significant differences compared with *rēng* 扔

Near-synonyms of <i>rēng</i> 扔	Conceptual feature	ESL speakers	HL speakers
<i>diū</i> 丢	Force	Y	N
	Speed	Y	N
	Direction	N	N
<i>pāo</i> 抛	Force	N	N
	Speed	N	N
	Direction	N	N
<i>shuāi</i> 摔	Force	Y	Y
	Speed	Y	Y
	Direction	Y	N

TABLE 3 Highlight of significant differences (*cont.*)

Near-synonyms of <i>rēng</i> 扔	Conceptual feature	ESL speakers	HL speakers
<i>shuǎi</i> 甩	Force	Y	N
	Speed	Y	N
	Direction	N	N
<i>tóu</i> 投	Force	N	N
	Speed	N	N
	Direction	N	N

Y = yes, there was a significant difference from *rēng* 扔; N = there was no significant difference

These results suggest that the HL speakers used the force, speed, or direction of near-synonyms in the similar way as *rēng* 扔, largely to the same degree as ESL speakers.

4.7 *Similarities to English throw*

A multivariate analysis of variance (MANOVA) was conducted comparing the English default throw verb throw to each of the Mandarin near-synonyms for HL speakers only. The MANOVA results showed that throw was not significantly different from *rēng* 扔 in all three conceptual features, which shows that both throw and *rēng* 扔 are generic verbs that represent the basic throwing action in each language. In other words, compared with the enactments of the five Mandarin near-synonyms (other than *rēng* 扔), HL speakers’ pantomime of throw was different. That is, HL speakers were not simply using the English word throw as a translation equivalent for Mandarin near-synonyms.

5 Discussion

We hypothesized that HL speakers would show signs of simplifying near-synonyms in their first and weaker language, Mandarin. Studies show that learners of English who already speak English at an advanced level still face challenges in accurate use of near-synonyms (Khazaal 2019; Majed 2017; Nada 2014; Shen 2023). Acquiring the ability to discriminate among synonyms or near-synonyms is an acquisition task at the core of lexical semantics. However, contrary to our expectations, the results revealed more similarities than differences between the two language groups. When enacting throw verbs

in Mandarin, there were no significant distinctions observed in force, speed, and direction between ESL and HL speakers (see Figs. 1 and 2). Moreover, for the near-synonyms for which the ESL speakers distinguished the core features from the generic throw verb *rēng* 扔, the HL speakers generally did, too (see Table 3). The one exception was the verb *diū* 丢: while the ESL speakers enacted this verb with significantly different force and speed than *rēng* 扔, the HL speakers did not. One previous study found that *diū* 丢 is the closest near-synonym throw verb in terms of meaning to *rēng* 扔 (Gao, 2001a). In sum, the HL speakers exhibited few signs of simplification in their enactments of Mandarin near-synonyms for throw verbs.

Further evidence for the strong semantic performance of the HL speakers can be seen from the degree of similarity across the Mandarin near-synonyms. When assessing the semantic distance between *rēng* 扔 and its five near-synonyms, the utilization of corpus data (e.g., Peking University CCL Corpus and Academia Sinica Balanced Corpus of Modern Chinese) along with Gao's (2001a) specification system of near-synonyms reveals a specific order in the semantic distance of *rēng* 扔 from other throw verbs: *rēng* 扔 → *diū* 丢 → *tóu* 投 → *pāo* 抛 → *shuǎi* 甩 → *shuāi* 摔, which shows that the closest one to *rēng* 扔 is *diū* 丢 and the farthest one is *shuāi* 摔. In the course of this study, it was observed that enactments produced by the participants largely matched the established rank order (see, for example, Figs. 1 and 2). This result, too, indicates that HL speakers demonstrated a solid understanding of the meanings of Mandarin throw verbs.

Nonetheless, some small variations surfaced in how ESL and HL speakers executed the throw verbs. The differences are manifested largely in terms of how similar the generic verb *rēng* 扔 is to the other throw verbs (see Tables 2 and 3). The results show that HL speakers acted out force and speed significantly differently from *rēng* 扔 only in *shuāi* 摔, while ESL speakers performed all three conceptual features significantly differently in *shuāi* 摔, two conceptual features in *diū* 丢, and one in *shuǎi* 甩. In total, HL speakers showed two significant differences, force and speed, within one word, while ESL speakers showed seven significant differences within three words. Thus, while overall, the enactments were quite similar, there were some differences between the HL speakers and the ESL speakers in how they enacted Mandarin verbs.

With the current data, we cannot rule out the possibility that the differences between HL speakers and ESL speakers were due to semantic transfer from English. We found that the HL speakers generally enacted Mandarin verbs differently from the English verb throw. This result suggests that they differentiated the Mandarin verbs from the generic English throw verb. However, it is important to keep in mind that the English verb throw shares core

features with the generic Mandarin verb *rēng* 扔 (Gao & Wang, 2012; Gao et al., 2016). Moreover, English is rich in near-synonyms for throw (e.g., *toss*, *lob*, *fling*, *cast*). Future studies can test for possible transfer from other English near-synonym verbs.

The HL speakers' sensitivity to near-synonyms in Mandarin throw verbs differs markedly from the semantic knowledge of L2 learners observed in other studies. L2 learners commonly find synonyms challenging (Higa, 1963; Laufer, 1990; Nation, 2001). Part of the reason for that challenge is that L2 learners typically do not encounter many synonyms or near-synonyms during the initial stages of their vocabulary acquisition. Teachers often prioritize instructing students on words that convey novel information rather than those conveying information similar to what students already know (Webb, 2007). Specifically in the realm of near-synonyms of throw, one study found that when choosing an English label for specific throwing actions, L1 Mandarin ESL speakers chose the more generic label throw rather than *lob*, as the English monolinguals did (Nicoladis & Gao, 2021). These results support the conclusion that the L2 learners were simplifying the distinctions between English near-synonyms.

One possible interpretation of these results is that HL speakers had acquired the meaning of these near-synonyms in childhood and maintained that knowledge through adulthood. In support of that interpretation, Gao (2001a) reported that the Mandarin throw verbs used in this study are acquired at an early age. Furthermore, Gao (2020) argued that linguistic knowledge and cognitive understanding of human physical actions for L1 speakers develop in tandem during their early years of life. Still, other studies have shown that the acquisition of specialized language is notably influenced by its use in discourse (Frankenberg-Garcia, 2018; Wongkhan & Thienthong, 2021). It is likely that the participants in this study learned the distinctions between the near-synonyms in everyday discourse, including, say, following instructions for throwing items into waste baskets. Future studies can disentangle how early exposure and/or naturalistic exposure is important in learning the fine distinctions between near-synonyms.

This study focused only on near-synonyms of throw verbs. Future studies can test whether the high degree of sensitivity to semantic distinctions among HL speakers extends to other semantic domains. Some previous research has shown that verbs may not share a conceptual store across languages to the same extent as concrete nouns (Van Hell & De Groot, 1998). So it is possible that HL speakers may be particularly likely to show sensitivity to language-specific meanings in verbs. In this study, we also tested participants' knowledge of single words with no other linguistic context. Future studies can test whether we observe the same level of sensitivity among HL speakers when the words are embedded in a linguistic context.

6 Conclusion

The purpose of this study was to test if HL Mandarin speakers simplified near-synonyms of Mandarin throw verbs. In the experiment, English-dominant HL speakers and Mandarin-dominant ESL speakers were asked to act out six Mandarin throw verbs. We compared the HL speakers’ enactments in terms of force, speed, and direction to those of ESL speakers. We found little evidence for simplification. Both groups of bilinguals performed similarly, regardless of their dominant language. We have interpreted these results as showing the positive effects of early and naturalistic exposure to Mandarin in learning near-synonyms. Future studies are needed to test whether it was the early exposure or the naturalistic exposure (or both) that led to the HL speakers’ sensitivity to the differences in near-synonyms. Future studies can also test whether these results generalize to other semantic domains. As the HL speakers were less proficient in Mandarin than the ESL speakers, the results of this study support previous research demonstrating that factors beyond proficiency have the capacity to influence access to meaning (Athanasopoulos et al., 2015; de la Riva López et al., 2012).

Appendix 1: Number of Participants Who Threw in Each Direction

Near-synonyms	Action	HL speakers	ESL speakers
<i>diū</i> 丢	Forward	10	6
	Upward	2	3
	Downward	8	12
	Sideways	1	2
<i>pāo</i> 抛	Forward	7	4
	Upward	10	17
	Downward	2	1
	Sideways	1	0
<i>rēng</i> 扔	Forward	14	14
	Upward	6	1
	Downward	1	5
	Sideways	1	1

(cont.)

Near-synonyms	Action	HL speakers	ESL speakers
<i>shuāi</i> 摔	Upward	1	0
	Downward	18	22
<i>shuǎi</i> 甩	Forward	4	5
	Upward	1	2
	Downward	4	8
	Backward	0	2
	Sideways	7	4
	Circular	4	2
<i>tóu</i> 投	Forward	8	9
	Upward	7	5

Notes on Contributors

Huong Hoang worked on this project when she was an undergraduate student (psychology major) at the University of Alberta.

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Helena Hong Gao did most of the work on this project as a faculty member at the Nanyang Technological University. Her research focuses on lexical semantics, particularly in bilingual populations.

Yu (Karen) Du worked on this project as a graduate student at the University of Alberta. She is currently working as a Post-Doctoral Fellow at the University of Arizona.

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